James Cenawood

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EDUCATION

Cornell University | College of Engineering, Ithaca, NY — Aug 2024 - May 2028 (Expected)

Bachelor of Science in Computer Science | GPA: 3.9

Coursework: Object-Oriented Programming & Data Structures, Functional Programming
In progress: Analysis of Algorithms, Machine Learning

EXPERIENCE

Northwell Health | Data Science Intern - New Hyde Park, NY — Jun 2026 - Aug 2026

- Built a natural-language-to-SQL agent in Python and LangGraph that validates structured query plans against Epic EHR schema metadata before compiling SQL; enforced deterministic safety checks without patient-data access
- Raised top-5 retrieval hit rate from 53.8% to 75% across ~250 analyst queries by combining BM25/SQLite full-text search with dense-vector retrieval over 40,551 Epic schema files and 482,130 chunks
- Tied for first in the intern Kaggle competition using an Optuna-tuned CatBoost and TabNet ensemble

Cornell Physical Intelligence | Software Subteam Lead - Ithaca, NY — Dec 2025 - Present

- Lead an eight-engineer subteam, owning the research agenda and implementation for mapless visual navigation and collective-thrust/body-rate control in Anduril's AI Grand Prix; passed Virtual Qualifier 1 at 17.2s
- Enforced race-time telemetry limits to block policies from reading unavailable state; built a Python/PowerShell harness for simulator launch, scoring, telemetry capture, and experiment summaries
- Designed and administered 20+ technical interviews using an original drone-themed algorithms problem set

Cornell Extended Reality | Software Team Member - Ithaca, NY — Aug 2025 - Present

- Designed a typed, multi-tenant memory backend for an AI smart-glasses assistant (FastAPI, PostgreSQL, SQLAlchemy), storing conversation episodes, time-scoped facts, summaries, and relationship graphs
- Built a natural-language person resolver with multi-pass name matching and disambiguation hints for descriptions that matched multiple stored identities

PROJECTS

AlgoSplit | Full-Stack Algorithmic Training Planner — 2025 - 2026

- Shipped authenticated web and iOS clients for an algorithmic training planner (FastAPI, Supabase, React, Expo), with workout logging, history, split comparison, and 3D stimulus visualization
- Cut p95 analysis latency from 31.9 ms to 3.85 ms (8.3x) in a local benchmark by porting the analysis kernel to Rust; checked numeric parity with Python within 1e-8
- Modeled training stimulus and fatigue across 29 muscle regions from hypertrophy research; increased model-scored net weekly stimulus by 35% on average across 100+ user-submitted splits

Surpassing the Leader | Exact Stochastic-Game Solver — 2026

- Built a stochastic-game solver and verification pipeline in Python, OCaml, C++, and Rust; proved a state-space reduction from 5.27 billion states to 289.4 million equivalence classes
- Replaced a linear-programming inner loop with a closed-form recurrence; solved the 289-million-class game in 49 seconds on a laptop, down from 1.5 hours on a 12-core desktop, with independent scalar verification

Sentinel | 1st Place, URMC x WICC x Gemini Hackathon — 2026

- Built a QR-based waiting-room triage prototype (TypeScript, LangChain, Gemini API) with conversational intake and a deterministic Emergency Severity Index (ESI) rules engine for a live patient queue
- Designed queue controls with emergency overrides and promotion hysteresis; constrained Gemini to advisory reordering within severity bands and escalation requests

SKILLS & INTERESTS

Languages: Python, SQL, TypeScript, Rust, C++, OCaml
Tools & Frameworks: FastAPI, PostgreSQL, SQLAlchemy, React, Supabase, Docker, Git, LangGraph, PyTorch, TensorFlow
Interests: Tournament Poker, Game Design, 3D Animation, Rock Climbing